THE RECOVERY DESK FOR INDIA'S MICRO AND SMALL SUPPLIERS

PayHaq

The agent that gets MSMEs paid.

Upload a ledger. PayHaq reconciles payments, finds invoices past the 45-day limit, computes statutory interest, audits the evidence, drafts the notice and Samadhaan claim, and follows through until cash arrives — with a human approving every send.

THE RECOVERY LOOP

01 **Detect**
45-day legal trigger, not a generic aging bucket.

02 **Prove**
Invoice, PO, challan, e-way bill and bank credit in one evidence graph.

03 **File**
Cited demand notice and Samadhaan claim, approved by the owner.

04 **Recover**
Follow-ups, response tracking, escalation and settlement.

45 days
the legal payment window

3×
the RBI bank rate, compounded

1 agent
from detection to recovery

TEAM
Update

MEMBERS
Shikhar Goel
Ishpreet Singh

INSTITUTION
Thapar Institute of Engineering and Technology, Patiala

SUBMISSION
3 November 2026
Round 1 · The Plan

01 / THE PROBLEM

Visibility without action.

India gave micro and small suppliers a legal remedy for late payments. Most never use it.

45 days

THE LEGAL PAYMENT WINDOW FOR MICRO AND SMALL SUPPLIERS UNDER SECTION 15 OF THE MSMED ACT, 2006.

3×

THE RBI BANK RATE, COMPOUNDED MONTHLY, THAT A DELAYED BUYER OWES UNDER SECTION 16.

A supplier can prove the invoice, the delivery and the delay. What they lack is the time to reconcile payments, the evidence discipline to build a claim, and the legal support to file it. So the aging report stays a report.

WHAT IT COSTS TODAY

- Working capital locked in receivables; payroll and supplier payments wait.
- Statutory interest never claimed — the buyer gets an interest-free loan.
- Invoices discounted at 2–4% a month to get cash sooner.
- The relationship still strains, because the chase is manual and emotional.

WHY EXISTING TOOLS STOP SHORT

TOOL	WHAT IT DOES	WHERE IT STOPS
Tally / Zoho Books	Shows aging and reminders	No evidence audit, no legal claim, no filing
Lawyer	Sends a notice	₹10–25k upfront; slow for ₹2–5L claims
Collection agency	Chases the buyer	Commission, pressure tactics, relationship damage
MSME Samadhaan	Accepts a delayed-payment application	Drafting, evidence and reconciliation are manual
PayHaq	Executes the recovery playbook	Detection → evidence → notice → filing → follow-up

THE PROBLEM STATEMENT

Indian micro and small enterprises can legally demand payment within 45 days and compound interest for delays, but most lack the time, evidence discipline, and legal support to convert an overdue invoice into a recovery action.

WHO HOLDS THE PROBLEM

- **Micro and small B2B suppliers** — 5–50 employees, ₹50L–₹5Cr turnover.
- **Owner-operators** selling to larger buyers on 60–120 day cycles.
- **CA / AR managers** handling 20–100 MSME clients.
- **Cluster associations** whose members face the same buyers and the same delays.

Early wedge: precision manufacturing clusters in Ludhiana, Pune, Coimbatore and Rajkot — dense, document-heavy, and used to 60–120 day terms.

SCOPE DISCIPLINE

WHAT WE TAKE ON

- B2B invoices from micro and small suppliers
- 45+ day delays and evidence gaps
- Statutory interest, demand notice and Samadhaan claim
- Follow-up, escalation and recovery tracking

WHAT WE DO NOT

- × Consumer debt or legal representation
- × Arbitration strategy or recovery from insolvent buyers
- × Lending, credit or invoice financing

Sources: MSMED Act, 2006, ss. 15, 16 and 18; MSME Samadhaan portal; RBI bank rate. Interest figures in this document are illustrative, not a promise of recovery.

THE 45-DAY CLOCK · WHAT THE LAW ALREADY PROVIDES

DAY 0
Invoice accepted
The supplier delivers goods or services; the 45-day clock starts.

DAY 45
Payment due
The buyer must pay. No payment means the statutory remedy begins.

DAY 46+
Interest accrues
Compound interest at 3× the RBI bank rate, compounded monthly.

NEXT
Samadhaan application
The supplier can file a delayed-payment application online.

IF UNPAID
Facilitation Council
The state MSEFC can conciliate and award interest and costs.

THE TAKEAWAY

"The remedy already exists. The execution layer does not."

02 / USERS

A finance team that does not exist.

The owner is the finance team, the legal team and the sales team. Recovery is the work that never gets a slot.

PRIMARY PERSONA

MICRO MANUFACTURER

Rakesh, 38

Owner, precision components unit · Pune

TEAM	REVENUE	BUYERS
22 employees	**₹3.2Cr / year**	**6 active**

PAYMENT CYCLE	OVERDUE	FINANCE / LEGAL
60–120 days	**₹38L**	**None in-house**

"I know who owes me. I don't have the time to chase legally, and I can't afford to lose the buyer."

JOBS TO BE DONE

- Know exactly who owes what, and which invoices are legally actionable.
- Preserve the buyer relationship while applying pressure.
- Act before the leverage disappears.
- Avoid upfront legal costs on a ₹2–5L claim.
- Prove the claim with documents the buyer cannot dispute.

SECONDARY PERSONA

Meera, 34 – CA

60 MSME clients · Pune

Wants a repeatable recovery workflow, not another inbox. Cannot chase for every client, but sees advisory revenue in solving it. PayHaq gives her a portfolio view and a defensible process.

WHAT SUCCESS LOOKS LIKE

Cash in the bank, not another report. A notice the buyer takes seriously, backed by evidence the owner already has.

ADOPTION WEDGE

Start with one cluster and one workflow: micro-manufacturers with 3–15 buyers and 20–100 open invoices.

A MONDAY IN RAKESH'S LIFE

09:00	11:00	14:00	16:00	18:00
Opens the aging report. Four buyers are past 60 days.	Calls the largest buyer. "Next month" again.	CA says GST and bookkeeping are in scope; recovery is not.	Lawyer quotes ₹25,000 upfront for one notice.	Promises himself he will "handle it next week."

FROM JOBS TO PRODUCT DECISIONS

WHAT THE OWNER NEEDS	WHAT PAYHAQ DOES	WHERE IT LIVES
Know who owes what	Reconciled case queue ranked by value, age and evidence score	Reconciler + dashboard
Preserve the relationship	Tone control: relationship-preserving or firm; every send human-approved	Legal Drafter + approval gate
Act before leverage disappears	45-day detection, deadline tracking and next-best-action prompts	Tracker
Avoid upfront legal costs	A fixed workflow that produces the first notice without a lawyer retainer	PayHaq
Prove the claim	Evidence graph, missing-document flags and section-level citations	Evidence Auditor + legal RAG

THE TAKEAWAY

"The buyer holds the leverage until the supplier has a process."

03 / THE SOLUTION

An agent that executes recovery.

PayHaq is the action layer between an aging report and a recovered invoice.

ONE LINE

Upload your ledger. PayHaq reconciles payments, finds invoices past 45 days, computes statutory interest, audits the evidence, drafts the notice and Samadhaan claim, files it, and follows up — with you approving every send.

WHY AN AGENT IS REQUIRED

01

Multi-step, multi-week

Reconcile → audit → draft → file → track. A single prompt cannot own a workflow that spans weeks.

02

Real tool use

Ledger parser, OCR, interest calculator, legal retrieval, document generator and portal automation.

03

Memory that matters

Buyer history, past promises, correspondence and evidence status — remembered across cases.

04

Human-in-the-loop

Irreversible actions need approval. Every step is logged, reviewable and reversible before send.

THE ORIGINAL ANGLE

Every claim carries an **evidence graph**. Each amount and each legal assertion is traceable to a document, a section of the MSMED Act, and a timestamp. The agent cannot send a notice it cannot prove.

BEFORE / AFTER

TODAY	WITH PAYHAQ
Aging report	Actionable case queue
Manual calls	Agent-drafted, cited notice
Lawyer quote	12-minute packet
Weeks of delay	Same-day send
No evidence trail	Evidence score + audit log

DIFFERENTIATION

PLAYER	WHAT THEY OWN	WHAT THEY MISS
Tally / Zoho	Visibility	Action
Lawyer	Advice, notices	Cost, speed
Collection agency	Pressure	Relationship, transparency
Samadhaan	Filing window	Drafting engine
PayHaq	End-to-end execution	—

WHY NOW

FORMALISATION

Udyam registration gives a verifiable micro/small status — the trigger for the 45-day rule and the interest claim.

DIGITAL EVIDENCE

GST invoices, e-way bills and bank statements make the claim provable without a paper chase.

A FILING ROUTE EXISTS

Samadhaan and state Facilitation Councils exist. The missing layer is drafting, evidence and follow-through.

PRODUCT PRINCIPLES

Evidence before action. Human before send. Cash before dashboards.

THE TAKEAWAY

"PayHaq turns a report into a recovery action."

04 / PRODUCT WORKFLOW

From ledger to recovery.

Nine steps, five agents, three human gates.

01
INGEST & RECONCILE
Meet the owner where the data already is.
- CSV / XLSX or Tally export
- Match invoices to payments
- Handle partials and credit notes
- Flag ambiguous matches

AGENT · RECONCILER

›

02
DETECT & AUDIT
Turn an aging bucket into a legal case.
- 45+ days, micro/small only
- PO, invoice, challan, e-way bill
- Bank credit and correspondence
- Evidence completeness score

AGENT · EVIDENCE AUDITOR

›

03
QUANTIFY & DRAFT
Make the claim specific and defensible.
- Principal + statutory interest
- 3× RBI bank rate, monthly
- Cited demand notice
- Samadhaan claim statement

AGENT · LEGAL DRAFTER

›

04
APPROVE & FILE
The owner stays in control.
- Review every notice
- Choose tone: preserve or firm
- Send email / registered post
- Prepare Samadhaan packet

HUMAN GATE · APPROVE

›

05
TRACK & RECOVER
Recovery is a follow-through game.
- Follow-up cadence
- Classify buyer responses
- Escalate to Facilitation Council
- Track settlement and cash

AGENT · TRACKER

WORKED SCENARIO

Sharma Precision Components · Pune

12 MINUTES

A 214-row ledger, four buyers, no in-house finance team.

₹38.4L
principal across 9 overdue invoices

≈₹3.2L
statutory interest, illustrative

4 + 1
notices + Samadhaan claim drafted

Two missing delivery challans are flagged before drafting. The owner approves; the notices go out. The same work would have taken two days and a lawyer quote.

AGENT ROLES

AGENT	INPUT	OUTPUT
Reconciler	Ledger, bank statement	Matched invoices, exceptions
Evidence Auditor	Invoice, PO, challan, e-way bill	Completeness score, missing docs
Legal Drafter	Evidence pack, legal corpus	Cited notice + claim
Filing Agent	Approved packet	Sent notice / application
Tracker	Responses, deadlines	Next action, escalation

HUMAN GATES

Approve notice · Approve filing · Approve settlement. The agent never sends without a human yes.

TWO TONES, SAME FACTS

RELATIONSHIP-PRESERVING

Subject: Payment against Invoice #SP-1042

"We value the partnership and want to resolve this quickly. Our records show Invoice #SP-1042 for ₹4,80,000 is now 62 days past due. Could we agree a payment date this week? If there is a dispute, tell us what is missing and we will fix it."

FIRM

Subject: Statutory demand — Invoice #SP-1042

"Despite follow-ups, Invoice #SP-1042 for ₹4,80,000 remains unpaid 62 days after acceptance. Under Sections 15 and 16 of the MSMED Act, 2006, interest at 3× the RBI bank rate is now accruing. We request payment within 15 days to avoid a Samadhaan reference."

Same evidence. Same claim. The owner chooses the pressure.

THE TAKEAWAY

"Every step is evidence. Every irreversible step is human."

05 / FEATURES

A narrow MVP that runs end-to-end.

Round 2 proves the core workflow. Round 3 deepens it. Nothing is decorative.

MVP · ROUND 2

FEATURE	WHY IT MATTERS	IN THE BUILD
Ledger ingestion	Meets the owner where the data already is	CSV / XLSX + Tally export
Reconciliation	Partial payments and credit notes break naive claims	Deterministic matching + exceptions
45-day detection	The legal trigger, not a generic aging bucket	Micro/small filter + invoice age
Interest engine	The number that makes a claim credible	3× RBI bank rate, monthly compounding
Evidence audit	Missing documents are why claims fail	Checklist + completeness score
Notice + claim	The artifact the buyer and portal see	PDF templates + section citations
Dashboard + audit	Trust requires visibility and a trail	Pipeline, approvals, event log
Mock Samadhaan filing	End-to-end without portal risk	Downloadable packet

ROUND 3 · THE THREE IMPROVEMENTS

1 **Live Samadhaan filing**
Playwright automation with human confirmation and a downloadable fallback packet.

2 **Buyer correspondence agent**
Email/WhatsApp follow-ups, response classification, settlement options, tone control.

3 **CA multi-client workspace**
Portfolio view, bulk actions and client-level recovery reports.

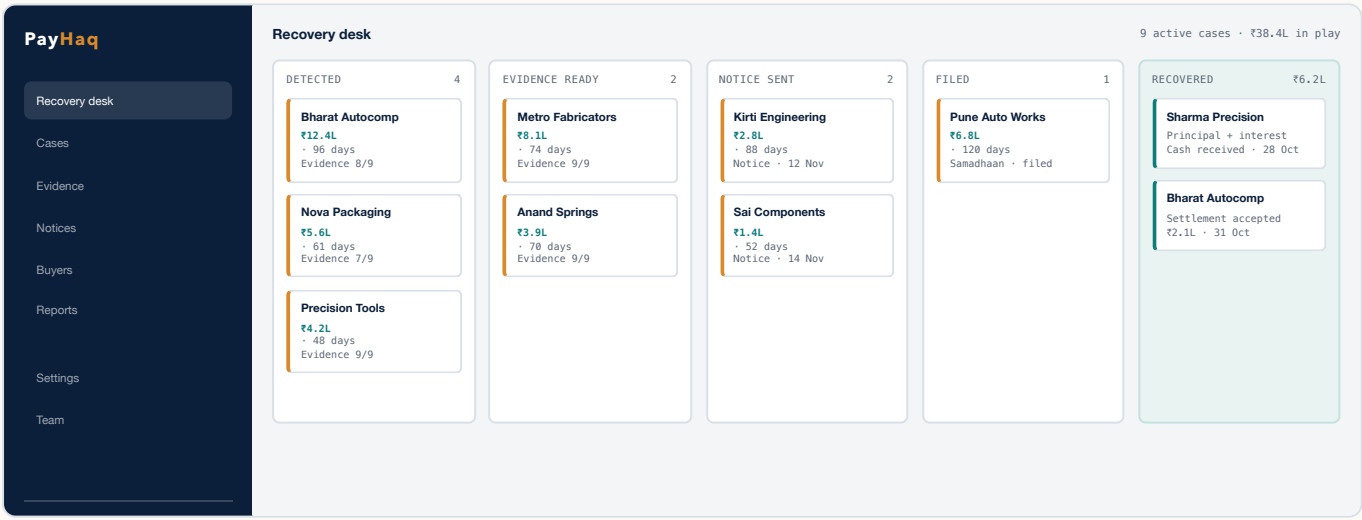
AFTER THE EVENT

Tally/Zoho native integrations · TReDS and invoice financing · state Facilitation Council packs · Hindi, Punjabi and Marathi.

WHAT WE WILL NOT BUILD

Legal representation · consumer debt · lending · arbitration strategy. **Scope is a feature.**

THE RECOVERY DESK · PRODUCT SURFACE



THE 3-MINUTE ROUND 2 DEMO

01 **Upload**
Drop a 214-row ledger exported from Tally. No cleanup.

02 **Review**
See 9 overdue invoices surface, with evidence gaps and statutory interest.

03 **Approve**
Approve a cited notice and download the Samadhaan-ready claim packet.

04 **Recover**
Track the response, the deadline and the cash on one dashboard.

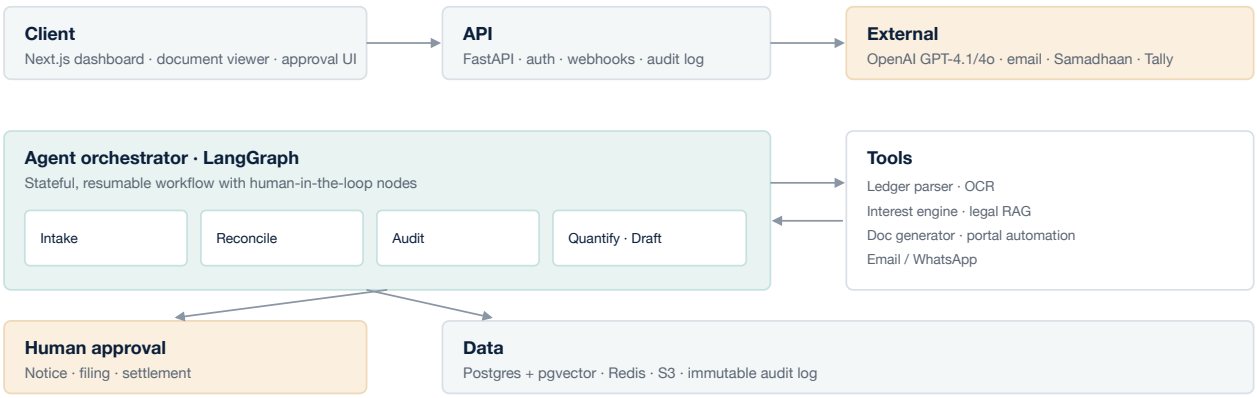
THE TAKEAWAY

"The demo is the workflow, not a mock."

06 / TECHNICAL PLAN

Reliable by design.

The hard part is not the chat. It is proving the claim.



STACK · AND WHY

LAYER	CHOICE	WHY
Frontend	Next.js · TypeScript · Tailwind · shadcn/ui	Fast dashboard, typed contracts, clean document viewer
Backend	FastAPI · Python 3.12	Ledger/document libraries, typed schemas, async jobs
Agent	LangGraph · OpenAI GPT-4.1/4o	Stateful, resumable workflow with human-in-the-loop nodes
Data	Postgres + pgvector · Redis · S3	One datastore for cases, evidence and legal retrieval
Documents	React-PDF / WeasyPrint	Versioned notice and claim templates
Observability	Langfuse · Sentry	Trace every agent step; catch failures before the demo

RELIABILITY BY DESIGN

Deterministic interest engine

Pure Python + Decimal. 3× RBI bank rate, monthly compounding. Unit tests against worked examples. No LLM arithmetic.

Legal RAG with citations

MSMED Act, Samadhaan guidelines, RBI notifications. Section-level metadata. “No citation, no claim.” Last-verified date on every rule.

Evidence graph

Invoice → PO → challan → e-way bill → bank credit. A missing link becomes an evidence gap, not a guess.

HARDEST PROBLEM

Messy ledgers and legal accuracy. Deterministic matching + confidence scores; the user resolves ambiguous matches; human approval at every irreversible step.

EVALUATION HARNESS

50 synthetic ledgers with known ground truth. Metrics: reconciliation accuracy, overdue recall, interest exactness, citation correctness, notice completeness, time-to-notice. Security: encryption at rest and in transit, role-based access, data minimisation, no training on user data.

LATENCY BUDGET

STEP	TARGET
Ledger parse	< 30 seconds
Reconcile + exceptions	< 60 seconds
Evidence audit	< 2 minutes
Draft notice + claim	< 30 seconds
Human review	As long as the owner needs
Samadhaan filing · Round 3	< 5 minutes, with confirmation

GUARDRAILS

- **No citation, no claim.** Every legal assertion points to a source.
- **No approval, no send.** The agent cannot email, file or settle alone.
- **No LLM math.** Interest is computed by a deterministic engine.
- **No training on user data.** Documents stay inside the tenant.
- **Every action is audited.** Who saw what, when, and what changed.

07 / DEVELOPMENT PLAN & IMPACT

A 72-hour MVP. A real path after.

Scope we can finish scores higher than scope that sounds impressive.

ROUND 1 · 2–3 NOV

The plan

User research, legal source pack, architecture, this PDF.

OWNER

Both

ROUND 2 · 4–7 NOV

The build

Ledger ingestion, reconciliation, interest engine, evidence audit, notice/claim PDF, dashboard, approval + audit, mock filing.

OWNER

Shikhar: agent/backend/legal
Ishpreet: frontend/docs/deploy
Both: integration/demo

ROUND 3 · 10–13 NOV

The finish

Live Samadhaan filing, correspondence agent, CA workspace, evaluation, polish.

OWNER

Shikhar: filing/correspondence
Ishpreet: workspace/UI
Both: final demo

RISKS AND MITIGATIONS

RISK	MITIGATION
Legal accuracy	Section-cited RAG, deterministic rules, human review, legal advisor review before pilot
Messy ledgers	Deterministic matching + confidence scores; user resolves exceptions
Portal changes	Downloadable packet fallback; human-assisted filing; Playwright only with confirmation

TARGET IMPACT

<15 min
TIME TO FIRST NOTICE

>90%
EVIDENCE COMPLETENESS BEFORE FILING

6–10 hrs
SAVED PER CASE

20
MSME PILOT IN ONE CLUSTER

Measure: notices sent, buyer responses, settlements, ₹ recovered, and user trust (SUS). Cash flow, not dashboards.

AFTER THE HACKATHON

0–3 MONTHS

Pilot with 20 MSMEs through a cluster association. Legal advisor review. Case studies with recovery numbers.

3–6 MONTHS

Tally/Zoho integrations. TReDS and invoice-financing partnerships. State-specific Facilitation Council packs.

6–12 MONTHS

Local languages. CA seats and portfolio pricing. API for lenders and incubators.

THE 72-HOUR BUILD SEQUENCE

DAY 1 · DATA + MATH

Data model, CSV/XLSX ingestion, reconciliation, statutory interest engine, test fixtures.

Lead: Shikhar · **Support:** Ishpreet

DAY 2 · WORKFLOW

Evidence audit, legal RAG, notice/claim PDF, dashboard, approval flow and audit log.

Lead: Ishpreet · **Support:** Shikhar

DAY 3 · DEMO

Mock filing, deployment, 50-case evaluation, 3-minute demo script, updated PDF.

Lead: Both

WHY THIS TEAM CAN SHIP IT

Two builders, complementary roles, a narrow wedge, and a 72-hour plan that ends with a deployed product — not a slide.

THE TAKEAWAY

"A narrow scope is a credible promise."

08 / TEAM

Team Update.

Two builders, one problem, and a workflow we can demo end-to-end.

SG

Shikhar Goel

PRODUCT & AI LEAD

Owns the problem research, the agent's decision model and the evidence behind every claim. Built the legal retrieval and statutory interest logic for PayHaq.

FOCUS

Product research · agent orchestration · legal RAG · statutory interest engine · evaluation

IS

Ishpreet Singh

FULL-STACK & DESIGN LEAD

Owns the product surface, the document pipeline and the demo experience. Built the Next.js dashboard, notice/claim templates and deployment.

FOCUS

Next.js · TypeScript · UI/UX · document generation · deployment · demo

CERTIFICATE LINES · EACH MEMBER SUBMITS THEIR OWN ENTRY

Shikhar Goel — Product & AI Lead. Led product research and built the agent orchestration, legal retrieval and statutory interest engine.

Ishpreet Singh — Full-Stack & Design Lead. Built the Next.js dashboard, document pipeline and deployment, and designed the end-to-end user experience.

WHY WE WIN

- **Specific legal wedge:** 45 days and 3× RBI interest, not a generic “AI for MSMEs.”
- **Agentic execution:** the agent files and follows up, with human approval at every irreversible step.
- **Feasible demo:** a real ledger, a real notice, a real claim packet in 72 hours.

INSTITUTION

Thapar Institute of Engineering and Technology, Patiala

TEAM

Update · 2 members · CodeNexus India 2026

ROUND 2 OWNERSHIP · 4–7 NOV

WORKSTREAM	LEAD	DELIVERABLE BY 7 NOV
Ledger ingestion + reconciliation	Shikhar	CSV/XLSX + exception queue
Evidence audit + legal RAG	Shikhar	Cited notice + claim statement
Dashboard + document pipeline	Ishpreet	End-to-end UI
Deployment + demo	Ishpreet	Live link + 3-minute script
Integration + evaluation	Both	50-case test set + submission

HOW WE WORK

- Ship the workflow before the polish.
- Every irreversible step has a human gate.
- Demo the real ledger, not a mock.
- Cut scope, not the evidence.

WHAT WE WILL SHOW IN ROUND 2

A real ledger reconciled · evidence gaps surfaced · a cited notice and claim packet · the deployed dashboard.

“The money is already owed. PayHaq goes and gets it.”